

Statistical Learning and Machine Learning

Exercise Session (Week 12)

SOLUTIONS

NONLINEAR CLASSIFICATION

- 1) No, Perceptron is a linear classifier and it would misclassify around half of the data (from both classes)
- 2) No, Logistic Regression is a linear classifier and it would misclassify around half of the data (from both classes)
- 3) We can use SVM if we use a non-linear kernel function (e.g., RBF kernel or polynomial kernel).
- 4) No, Linear Regression is a linear classifier and it would misclassify around half of the data (from both classes)
- 5) One transformation is the distance to zero (or the l_2 -norm) transformation. This would map all red points close to zero and all blue points to values higher than one. Then a classifier is a threshold in between, e.g., 1.
- 6) This transformation would map all red points close to zero and all blue points to values higher than one. The SVM solution will also be a threshold value close to 1.
- 7) The support vectors are the data points (values) which are the closest to the threshold from both sides of the threshold (two classes).
- 8) Points $\mathbf{x}_5$, $\mathbf{x}_8$ and $\mathbf{x}_9$ will be classified to class-red (because their norm is smaller than one). Data points $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_4$, $\mathbf{x}_6$, $\mathbf{x}_{10}$ and $\mathbf{x}_{11}$ will be classified to class-blue (because their norm is bigger than one). Data points $\mathbf{x}_3$ and $\mathbf{x}_7$ may be classified to either of the classes.